

Babar Hashmi

AI Engineer — Machine Learning, R&D

LinkedIn

— GitHub

Bangalore, India

— Open to Relocation

Professional Summary

Results-driven **AI Engineer** with around 3 years of developing **machine learning solutions**, focusing on **deep learning**, **agentic AI systems**, and scalable **model deployment**. Proven track record delivering **\$50K+ cost savings**, **92% model accuracy**, and **40% faster time-to-production**. Expert in **custom agent engineering**, **LLM fine-tuning**, **MLOps**, and **end-to-end ML pipelines**.

Core Competencies

- **End-to-End MLOps**: Taking models from research to production-grade deployment and monitoring
- **Agentic AI System Design**: Building tool-calling agents and multi-step reasoning loops
- **Computer Vision Pipelines**: Real-time detection systems for physical-world use cases
- **LLM Fine-Tuning & Grounding**: Improving contextual accuracy and eliminating hallucination
- **SDK & API Engineering**: Designing developer-facing tools adopted across business units
- **Cross-Functional Delivery**: Partnering with stakeholders from research through internal rollout

Professional Experience

AI Engineer – R&D — QuillAI Network (QuillAudits)

Jul 2024 – Aug 2026

- **Architected and deployed** production **Agentic harness** for smart contract vulnerability detection, improving audit accuracy by **25%** and reducing manual review by **60%**
- **Engineered custom agentic AI systems** using the ADK/SDK — designing tool-calling loops and agent harness architecture; shipped developer-facing SDK across **3 business units**
- **Partnered with C-level stakeholders** on client-facing deployment and developer integration, delivering **predictive analytics** across business units
- **Pioneered audit harness** achieving **25% improvement** in contextual understanding, directly reducing customer escalations

AI/ML Researcher — QuillAudits

Sep 2023 – Jun 2024

- **Spearheaded development** of an AI-powered blockchain auditing tool as lead AI Engineer, reducing manual audit time by **30%**
- **Implemented statistical modelling** for blockchain security, processing **10K+ smart contracts** and identifying **95% of vulnerabilities**
- **Conducted advanced data analysis** for market due diligence, analysing **\$50M+ investments** using deep learning techniques
- **Drove Agile delivery** across cross-functional teams, improving delivery speed by **35%**

Key Projects & Achievements

GoldenHour

- **Shipped live full-stack app** guiding post-scam recovery via Next.js and TypeScript with middleware-enforced signed sessions.
- **Architected advisory LLM pipeline** with deterministic fallback; schema validation and timeouts ensure zero downtime if offline.

DesiRAG — Voice-Driven Hindi RAG

- **Built end-to-end RAG pipeline** featuring length-aware chunking, local embeddings, vector indexing, and hybrid RRF retrieval.
- **Implemented safety guardrails** to eliminate hallucinations and off-topic responses using benchmarked retrieval metrics.

Financial Strategy Risk Scoring Engine

Deterministic, Fact-Grounded Risk Verdicts for Real-World Investment Strategies

- **Built a deterministic scorecard engine** rendering risk verdicts for live investment strategies from real, point-in-time market data across **7 independent risk axes** — never a hand-written or model-generated verdict
- **Engineered a groundedness-gated LLM interface** where the model only routes intent and phrases output in plain language; a verifier rejects the entire response if any claim can't be traced to source data
- **Validated the engine against real historical market collapses** and shipped a one-command verification pipeline that regenerates and diffs every published number against committed evidence

Technical Proficiencies

- **ML/AI Frameworks**: TensorFlow, PyTorch, Transformers, Hugging Face, OpenAI SDK/ADK, custom agent frameworks, PoC RAG
- **Computer Vision**: YOLOv8, GroundingDINO, OpenCV, Vision-Language Models
- **Cloud & MLOps**: AWS, OCI, Docker, CI/CD pipelines, model monitoring
- **Engineering & Dev Tools**: TDD, agent harness design, SDK design, architecture review
- **Programming & Runtimes**: Python, TypeScript, Bun, Solidity, SQL/NoSQL, Pandas, NumPy, Matplotlib, Plotly, Streamlit

Certifications

- **Agentic AI Certified Associate** — Oracle *Jul 2026*
- **Copado Certified AI** — Copado *Jul 2026*
- **OCI: GenAI Certified Professional** — Oracle *Jul 2024*
- **Prompt & AI Engineering Certification** — MFT *Jun 2024*

Impact Snapshot

\$50K+ cost savings — 92% model accuracy — 25% LLM improvement — 60% faster manual review — Agentic AI systems delivered across 3 business units

Domains of Expertise

- **Product & R&D**: Led product engineering, product onboarding, R&D initiatives, and developer experience across AI-driven products
- **FinTech**: Deterministic, fact-grounded risk-scoring pipelines
- **Public Infrastructure**: Municipal-scale computer vision deployments
- **Blockchain & Web3 Security**: Smart contract auditing, vulnerability detection, on-chain analytics
- **Construction Tech**: Site inspection automation, MEP element detection